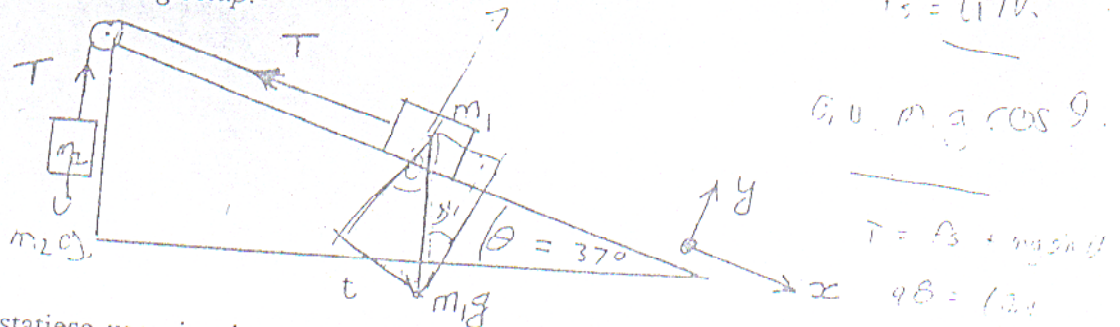


- (c) Beskou die volgende opstelling:
Consider the following setup:



Indien die statiese wrywingskoëffisiënt $\mu_s = 0,4$ is moet ons die waardes van m_2 bepaal wat die sisteem in rus sal hou.
If the coefficient of static friction is $\mu_s = 0,4$ we have to determine the range of values for m_2 to keep the system at rest.

- (i) As m_2 klein is, sal m_1 neig om af te gly. / With m_2 small, m_1 would tend to slide down.

Die som van die kragte in y-rigting op m_1 is: / The sum of the forces in y-direction on m_1 is:

$$N - m_1 g \cos \theta$$

(1)

Die som van die kragte in x-rigting op m_1 is: / The sum of the forces in x-direction on m_1 is:

$$-T - \mu_s N + m_1 g \sin \theta$$

- (ii) Spankrag T in die tou weens m_2 : / Tension in the rope due to m_2 :

(1)

$$T_{m_2} = m_2 g$$

- (iii) Die normaalkrag N op m_1 is: / The normal force N on m_1 is:

(1)

$$N = m_1 g \cos \theta$$

- (iv) Met $m_1 = 10$ kg, toon aan dat $m_2 = 2,8$ kg moet wees. / With $m_1 = 10$ kg, show that $m_2 = 2,8$ kg.

(1)

$$\begin{aligned} -T_{m_2} - \mu_s N + m_1 g \sin \theta + N - m_1 g \cos \theta &= 0 \\ -m_2 g - \mu_s (0,4 \cdot m_1 g \cos 37^\circ) + m_1 g \sin 37^\circ &= 0 \\ -m_2 g - \mu_s (0,4 \cdot 10 \cdot 9,8 \cdot \cos 37^\circ) + 10 \cdot 9,8 \cdot \sin 37^\circ &= 0 \\ -m_2 g - 31,3065 + 58,98 &= 0 \\ -m_2 g &= -26,67 \\ m_2 &= 2,72 \end{aligned}$$

(4)

Vraag / Question 5(c)(v)(vervolg / continue) / ...

$$\begin{aligned} -m_2 g - 31,3065 + 58,98 &= 0 \\ -m_2 g &= -26,67 \\ m_2 &= 2,72 \end{aligned}$$